MATH 440/540 **HW 11** Schreyer

Due Thurs. Oct 31

WITH SOLUTIONS

Please follow the homework instructions listed in HW 1 for writing up your homework. **When graphing the Fourier series representation of a function, be sure to graph at least 3 periods.**

1. (4 pts) Z& T Chapter 7, 8.7 Suppose that f is defined and continuous on $[-\pi, \pi]$. State the conditions under which the Fourier series of f converges to $f(x)$ for every x in $[-\pi, \pi]$.

   The conditions for point-wise convergence (the periodic extension of $f(x)$ and $f'(x)$ are piecewise continuous) gives us that the FS of $f(x)$ converges to $f(x)$, wherever the periodic extension of $f(x)$ is continuous. We have that $f(x)$ is continuous for $[-\pi, \pi]$. We need two more conditions: (1) $f'(x)$ is piecewise continuous (only finite jumps allowed at discontinuities) and (2) to guarantee that the FS converges to $f(x)$ we need no jumps in the periodic extension of $f(x)$, so we also need that $f(-\pi) = f(\pi)$.

2. (12 pts) Z& T Chapter 7, 8.10 a, c

   Derive the Fourier series for each of the following functions and then sketch the graph of the function to which the series converges for all x.

   (a)

$$f(x) = \begin{cases} 0 & \text{for } -\pi < x < 0 \\ 1 & \text{for } 0 < x < \pi \end{cases}$$

   (c)

$$f(x) = x^2, \qquad -\pi \le x \le \pi$$

   (a)

$$f(x) = \begin{cases} 0 & \text{for } -\pi < x < 0 \\ 1 & \text{for } 0 < x < \pi \end{cases}$$

   We note that this is neither an even nor odd function, so we must calculate all coefficients.

$$\begin{aligned} a_0 &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx = \frac{1}{\pi}\left[\int_{-\pi}^{0} f(x)\,dx + \int_{0}^{\pi} f(x)\,dx\right] \\ &= \frac{1}{\pi}\left[\int_{-\pi}^{0} (0)\,dx + \int_{0}^{\pi} (1)\,dx\right] = \frac{1}{\pi}\big[\pi\big] = 1. \end{aligned}$$

$$
\begin{aligned}
a_n &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(nx)\,dx = \frac{1}{\pi}\left[\int_{-\pi}^{0} f(x)\cos(nx)\,dx + \int_{0}^{\pi} f(x)\cos(nx)\,dx\right] \\
&= \frac{1}{\pi}\left[\int_{-\pi}^{0}(0)\,dx + \int_{0}^{\pi}\cos(nx)\,dx\right] \\
&= \frac{1}{n\pi}\sin(nx)\Big|_{x=0}^{\pi} = \frac{1}{n\pi}\big[0-0\big] = 0.
\end{aligned}
$$

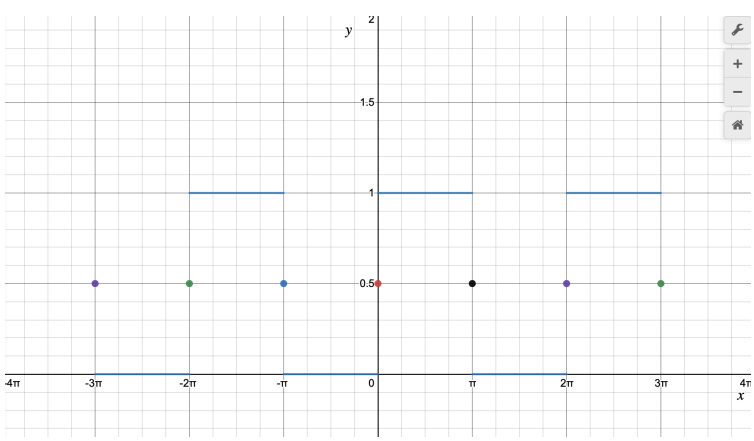
So why would these coefficients be zero? Looking at the graph (below) we see that after shifting vertically by $1/2$, we have an odd function ! so this is why $a_n = 0$.

$$
\begin{aligned}
b_n &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin(nx)\,dx = \frac{1}{\pi}\left[\int_{-\pi}^{0} f(x)\sin(nx)\,dx + \int_{0}^{\pi} f(x)\sin(nx)\,dx\right] \\
&= \frac{1}{\pi}\left[\int_{-\pi}^{0}(0)\,dx + \int_{0}^{\pi}\sin(nx)\,dx\right] \\
&= -\frac{1}{n\pi}\cos(nx)\Big|_{x=0}^{\pi} = -\frac{1}{n\pi}\left[(-1)^n - 1\right]
\end{aligned}
$$

Here we see that for n even, we have $b_n = 0$. So letting $n = 2k-1$ we have that the Fourier series is:

$$
f(x) = \frac{1}{2} + \frac{2}{\pi}\sum_{k=1}^{\infty}\frac{1}{2k-1}\sin[(2k-1)x]
$$

A sketch of the graph of the Fourier series of $f(x)$ made using Desmos is given by
Grader: 1 point off if 3 periods at least are not graphed.



(c)

$$
f(x) = x^2, \qquad -\pi \le x \le \pi
$$

Since this is an even function, we know that the FS of this function must also be even, and hence, $b_n = 0$.

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx = \frac{1}{\pi}\int_{-\pi}^{\pi} x^2\,dx = \frac{1}{3\pi}x^3\Big|_{x=-\pi}^{\pi} = \frac{1}{3\pi}\left[\pi^3 - (-\pi)^3\right] = \frac{2}{3}\pi^2$$

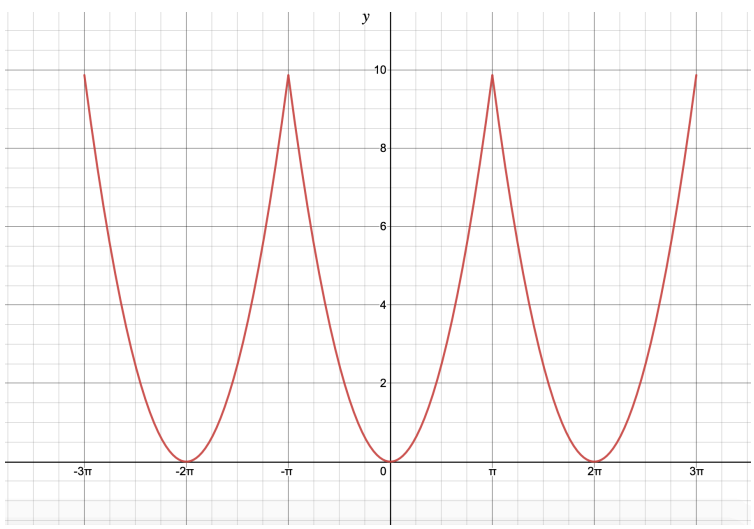
To integrate the following we integrate by parts twice. Not putting in all of the details we get:

$$\begin{aligned}
a_n &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(nx)\,dx = \frac{1}{\pi}\int_{-\pi}^{\pi} x^2\cos(nx)\,dx \\
&= \frac{1}{\pi}\left[\frac{n^2x^2-2}{n^3}\sin(nx) + \frac{2x}{n^2}\cos(nx)\right]_{x=-\pi}^{\pi} \\
&= \frac{1}{\pi}\left[\frac{n^2\pi^2-2}{n^3}\sin(n\pi) - \frac{n^2\pi^2-2}{n^3}\sin(n\pi) + \frac{2\pi}{n^2}\cos(n\pi) - \frac{-2\pi}{n^2}\cos(-n\pi)\right] \\
&= \frac{1}{\pi}\left[0 - 0 + \frac{4\pi}{n^2}(-1)^n\right] = \frac{4}{n^2}(-1)^n
\end{aligned}$$

Putting it all together, we have that the Fourier series:

$$f(x) = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}\cos(nx)$$

A sketch of the graph of the Fourier series of $f(x)$ made using Desmos is given by
Grader: 1 point off if 3 periods at least are not graphed.



3. (10 pts) Z& T Chapter 7, 8.10 b
   Derive the following Fourier series representation, and sketch the function to which it

converges:

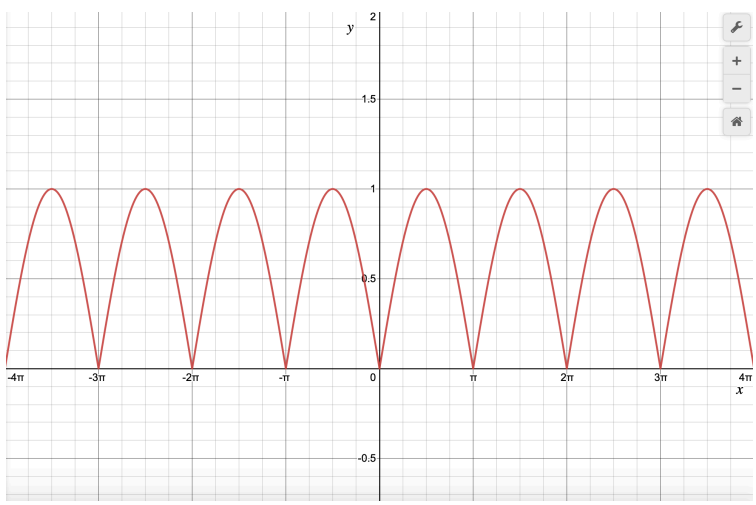
$$\sin(x) = \frac{2}{\pi} - \frac{4}{\pi}\sum_{k=1}^{\infty}\frac{1}{4k^2-1}\cos(2kx), \qquad 0 \le x \le \pi$$

At first this looks suspicious - we have an odd function on the right and an even function on the left. How can they be equal?! But the domain of the expression is only $0 \le x \le \pi$, and so we are looking at an even extension of $\sin(x)$! The even extension of $\sin(x)$ is $|\sin(x)$ which we did in the above problem, 7.7b. So for an even extension we have $b_n = 0$ and using the results of 7.7b we have

$$\begin{aligned} a_0 &= \frac{4}{\pi} \\ a_k &= \frac{4}{\pi}\frac{1}{1-4k^2} = -\frac{4}{\pi}\frac{1}{4k^2-1} \end{aligned}$$

where $k = 2n$. This gives us the desired result.

A sketch of the graph of the Fourier series of $f(x)$ made using Desmos is given by
Grader: 1 point off if 3 periods of the solution is not graphed.



4. (16 pts) Z& T Chapter 7, 8.9
Find the Fourier cosine and sine series of the function $f(x) = x(\pi - x)$, $0 \le x \le \pi$ and then sketch the graphs of the two functions to which the series converges for all x. You may use technology to help with integration (indicating what technology you use), but you should be prepared to evaluate e.g. $\int x \sin(nx)\, dx$ by hand on a test.

First we note that $f(x)$ is piecewise C^1 so the Fourier series converges. To find the Fourier cosine series, we extend the function evenly to the interval $[-\pi, \pi]$. Since it is an even function, we have $b_n = 0$. Using the result of 7.7c, so that we do not have to perform the integrals again, we get

$$\begin{aligned} a_0 &= \frac{1}{\pi}\int_{-\pi}^{\pi} |x|(\pi - |x|)\, dx = \frac{2}{\pi}\int_0^{\pi} (\pi x - x^2)\, dx = \frac{\pi^2}{3} \\ a_n &= \frac{1}{\pi}\int_{-\pi}^{\pi} |x|(\pi - |x|)\cos(nx)\, dx = \frac{2}{\pi}\int_0^{\pi} (\pi x - x^2)\cos(nx)\, dx \\ &= -\frac{2}{n^2}\left[1 + (-1)^n\right] \end{aligned}$$

and we have that the Fourier cosine series is

$$f(x) = \frac{\pi^2}{6} - 2\sum_{n=1}^{\infty} \frac{1 + (-1)^n}{n^2}\cos(nx)$$

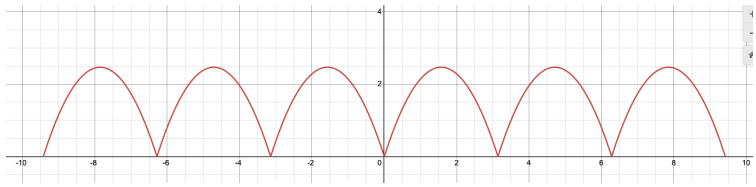
To find the Fourier sine series, we extend the function oddly to the interval $[-\pi, \pi]$. Since it is an odd function, we have $a_0 = a_n = 0$. We note that an odd function satisfies $f(-x) = -f(x)$. So on $[-\pi, 0]$ the extended function is $-f(x) = -x(\pi - x)$. Noting that an odd function (extended function) times an odd function ($\sin(nx)$) is even, we have (using Wolfram-Alpha technology to help):

$$\begin{aligned} b_n &= \frac{2}{\pi}\int_0^{\pi} x(\pi - x)\sin(nx)\, dx = \frac{2}{\pi n^3}\left(2 - 2(-1)^n\right) \\ &= \frac{4}{\pi}\frac{1 - (-1)^n}{n^3} \end{aligned}$$

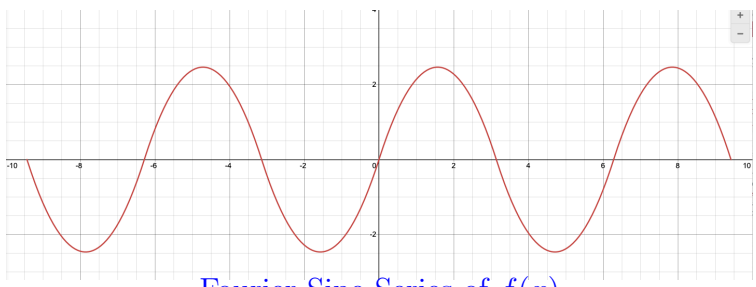
and we have that the Fourier sine series is

$$f(x) = \frac{4}{\pi}\sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n^3}\sin(nx)$$

These are graphed below. There are no discontinuities, so the F-S in each case converges to the function everywhere where $0 \le x \le \pi$.



Fourier Cosine Series of $f(x)$



Fourier Sine Series of $f(x)$

5. (10 pts) Z& T Chapter 7, 7.6
Consider the Dirichlet problem for the Laplace equation where Ω is the unit disk, $0 \leq r < 1$, $-\pi \leq \theta \leq \pi$, and the boundary condition given by $u(1, \theta) = |\theta|$, $-\pi \leq \theta \leq \pi$. The solution is derived in the text (you do NOT have to rederive the solution) and is given by

$$u(r, \theta) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2}{\pi} \frac{(-1)^n - 1}{n^2} r^n \cos(n\theta), \qquad 0 \leq r < 1.$$

Use the series solution to determine u at particular locations. Note that this solution is only valid for $r < 1$, so we would expect the series to converge more slowly as r approaches 1. Calculate u at the following locations to two decimal places (two digits after the decimal point) of accuracy, and note the number of *non-zero* terms that give us this solution.

(a) $u(0, \theta)$

(b) $u(0.1, 0)$

(c) $u(0.5, 0)$

(d) $u(0.9, 0)$

What is the value of the solution along the diameter, $\theta = \pm\pi/2$?

We calculate u at the following locations to two decimal places of accuracy, and note the number of *non-zero* terms that give us this solution.

(a) $u(0, \theta) = \frac{\pi}{2} \approx 1.57$ 1 term.

(b) $u(0.1, 0) \approx 1.44$ 2 terms.

(c) $u(0.5, 0) \approx .91$ 4 terms.

(d) $u(0.9, 0) \approx .26$ 14 terms.

We note that when $\theta = \pm\dfrac{\pi}{2}$, $\cos(n\theta) = 0(n = 1), -1(n = 2), 0(n = 3), 1(n = 4), ...,$ and so

$$\frac{(-1)^n - 1}{n^2} r^n \cos(n\theta) = 0$$

since the first term is 0 when n is even, and the second term is zero when n is odd. Thus

$$u\left(r, \pm\frac{\pi}{2}\right) = \frac{\pi}{2}.$$

6. (24 pts) Z& T Chapter 7, 7.7, a, b, c
Find the series solution to the following Dirichlet problem:

Find $u(r, \theta) \in C^2(\Omega) \cap C^0(\overline{\Omega})$ such that

$$\begin{aligned} &\triangle u(r, \theta) = 0; \qquad 0 \le r < 1, \quad -\pi \le \theta < \pi \\ &u(1, \theta) = f(\theta); \quad -\pi \le \theta < \pi \end{aligned}$$

where $f(\theta)$ is a given $C^0(\partial\Omega)$ function (check, mentally, that the following functions are $C^0(\partial\Omega)$):

(a) $f(\theta) = \theta^2$

(b) $f(\theta) = |\sin(\theta)|$

(c) $f(\theta) = |\theta|(\pi - |\theta|)$

You may use technology to help with integration (indicating what technology you use), but you should be prepared to evaluate e.g. $\int x \sin(nx)\, dx$ by hand on a test.

The solution is given by:

$$u(r, \theta) \quad = \quad \begin{cases} \frac{a_0}{2} + \sum_{n=1}^{\infty} r^n \left(a_n \cos(n\theta) + b_n \sin(n\theta)\right), & r < 1 \\ f(\theta), & r = 1 \end{cases}$$

where a_0, a_n, and b_n are the Fourier coefficients are of $f(\theta)$. So let us find the solutions for the following functions, $f(\theta)$, for $r < 1$:

Grader: No points off if there is nothing written about $f \in C^0(\partial\Omega)$.

(a) $f(\theta) = \theta^2$. We note that $f \in C^0(\partial\Omega)$ so the problem is well-posed. Using the fact that the integral of an even function over $[-\pi, \pi]$ is twice the area of the integral of the function over $[0, \pi]$ we get:

$$\begin{aligned}
a_0 &= \frac{1}{\pi}\int_{-\pi}^{\pi} \theta^2 \, d\theta = \frac{2}{\pi}\int_0^{\pi} \theta^2 \, d\theta = \frac{2}{3}\pi^2 \\
a_n &= \frac{1}{\pi}\int_{-\pi}^{\pi} \theta^2 \cos(n\theta) \, d\theta = \frac{2}{\pi}\int_0^{\pi} \theta^2 \cos(n\theta) \, d\theta = 4\frac{(-1)^n}{n^2} \\
b_n &= 0 \quad \text{since } f(\theta) \text{ is an even function}
\end{aligned}$$

The solution on the interior of the domain is thus

$$u(r, \theta) = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} r^n \cos(n\theta)$$

(b) $f(\theta) = |\sin(\theta)|$. We note that $f \in C^0(\partial\Omega)$ so the problem is well-posed. Using the fact that the integral of an even function over $[-\pi, \pi]$ is twice the area of the integral of the function over $[0, \pi]$ we get (using Wolfram-Alpha technology for evaluating the integral for a_n):

$$\begin{aligned}
a_0 &= \frac{1}{\pi}\int_{-\pi}^{\pi} |\sin(\theta)| \, d\theta = \frac{2}{\pi}\int_0^{\pi} \sin(\theta) \, d\theta = \frac{4}{\pi} \\
a_n &= \frac{1}{\pi}\int_{-\pi}^{\pi} |\sin(\theta)| \cos(n\theta) \, d\theta = \frac{2}{\pi}\int_0^{\pi} \sin(\theta)\cos(n\theta) \, d\theta \\
&= -\frac{1}{\pi}\left[\frac{(-1)^{n+1}}{1+n} + \frac{(-1)^{n+1}}{1-n} - \frac{1}{1+n} - \frac{1}{1-n}\right] \\
b_n &= 0 \quad \text{since } f(\theta) \text{ is an even function}
\end{aligned}$$

We can simplify the formula for a_n by noting that $a_n = 0$ when n is odd. Letting $n = 2k$ we get

$$a_k = \frac{4}{\pi}\frac{1}{1-4k^2}$$

We note that the coefficient k should be matched with $r^{2n}\cos(2k\theta)$ for consistency. Replacing k by n we then have

$$u(r, \theta) = \frac{2}{\pi} + \frac{4}{\pi}\sum_{n=1}^{\infty} \frac{1}{1-4n^2} r^{2n} \cos(2n\theta)$$

(the solution in the textbook writes the fraction in the series part of the solution as $4n^2 - 1$ with an appropriate sign change, probably so that the fraction is positive).

(c) $f(\theta) = |\theta|(\pi - |\theta|)$. We note that $f \in C^0(\partial\Omega)$ so the problem is well-posed. Using the fact that the integral of an even function over $[-\pi, \pi]$ is twice the area of the

integral of the function over $[0, \pi]$ we get:

$$\begin{aligned}
a_0 &= \frac{1}{\pi}\int_{-\pi}^{\pi} |\theta|(\pi - |\theta|)\, d\theta = \frac{2}{\pi}\int_0^{\pi} (\pi\theta - \theta^2)\, d\theta = \frac{\pi^2}{3} \\
a_n &= \frac{1}{\pi}\int_{-\pi}^{\pi} |\theta|(\pi - |\theta|)\cos(n\theta)\, d\theta = \frac{2}{\pi}\int_0^{\pi} (\pi\theta - \theta^2)\cos(n\theta)\, d\theta \\
&= (-1)^{n+1}\frac{2}{n^2} - \frac{2}{n^2} = \frac{2}{n^2}\left[-1 + (-1)^{n+1}\right] = -\frac{2}{n^2}\left[1 + (-1)^n\right] \\
b_n &= 0 \quad \text{since } f(\theta) \text{ is an even function}
\end{aligned}$$

The solution on the interior of the domain is thus

$$u(r, \theta) = \frac{\pi^2}{6} - 2\sum_{n=1}^{\infty} \frac{1 + (-1)^n}{n^2} r^n \cos(n\theta)$$